

Warren Olckers

BSc Computer Science | Machine Learning and AI | Data Analytics

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PROFILE

I am a BSc Computer Science graduate who spent the final stretch of my degree going deep into machine learning. My thesis benchmarked ML algorithms for network intrusion detection across three real-world datasets and built a reproducible XGBoost classifier, earning a grade of 1.0 (the highest distinction on the German grading scale). The oral defence was also graded 1.0. I have four years of experience doing supply chain and operational analytics in a live business environment, where data had to drive real purchasing, logistics, and cash flow decisions. I am comfortable working across the full data pipeline, from writing SQL queries and building data models through to producing dashboards and translating findings into clear recommendations for decision-makers.

KEY SKILLS

Programming: Python (OOP and functional), SQL, Java, JavaScript, Node.js, HTML/CSS

Machine learning: Supervised classification, XGBoost, feature engineering, model evaluation (precision, recall, F1, AUC), benchmark datasets (NSL-KDD, CICIDS2017, UNSW-NB15)

Data engineering: MySQL, PostgreSQL, MongoDB, data pipeline design, ETL processes, snowflake schema, dimensional modelling, data warehousing, data quality and validation

Analytics and BI: Power BI (dashboard development, interactive reports, DAX measures), data profiling, trend analysis, variance analysis, KPI reporting, data storytelling, stakeholder reporting

Mathematics: Probability, descriptive and inferential statistics, algorithms and data structures (grade 1.0 in Mathematics I)

Cloud and big data: Big Data Technologies, Cloud Computing (AWS and Azure fundamentals from coursework)

Tools: VS Code, Git, Excel (advanced), generative AI tooling, Power BI, CorelDRAW

Soft skills: Independent research, analytical rigour, stakeholder engagement, written and verbal communication (English C1, Afrikaans native, Dutch B1)

WORK EXPERIENCE

IT Analyst | Gummy Berry Juice South Africa | *December 2022 to present*

- Administered online sales channels and marketplace platforms, ensuring accurate product information, inventory management, pricing, order processing, and data integrity.
- Maintained and analysed customer, sales, inventory and supplier trends.
- Produced operational reports and analytical insights using Microsoft Excel and related reporting tools.
- Analysed inventory levels, stock trends, supplier performance, and logistics data to support purchasing decisions, identify risks, and improve operational efficiency.
- Monitored customer and supplier payment trends to support cash-flow planning and business performance analysis.
- Identified operational inefficiencies through data analysis and contributed to process improvement initiatives across multiple business functions.
- Provided first-line IT support for hardware, software, printers, and office technology systems, ensuring minimal disruption to business operations.
- Installed, configured, maintained, and troubleshooted computer systems, peripheral devices, and business technology infrastructure.
- Diagnosed and resolved technical issues to improve system reliability, operational continuity, and user productivity.
- Supported digital business operations through effective management of information systems, databases, and technology resources.
- Collaborated across operational, sales, and administrative functions to improve data accuracy, reporting quality, and business efficiency.

Founder and Operator | Custom Print Side Business | *Ongoing*

- Managed the full production and fulfilment cycle independently, using CorelDRAW and Canva to design and produce custom products, and handling all customer and order management.

RESEARCH AND PROJECTS

Bachelor's Thesis: ML in Network Intrusion Detection | | Completed 2026

Grade: 1.0 (highest distinction). Colloquium defence: 1.0.

- Conducted a systematic literature review of machine learning approaches applied to network intrusion detection systems, covering over 60 papers across multiple algorithm families.
- Compared algorithm performance across NSL-KDD, CICIDS2017, and UNSW-NB15 using accuracy, precision, recall, F1-score, and AUC as evaluation metrics.
- Designed and implemented a reproducible XGBoost baseline on the CICIDS2017 dataset, including feature engineering, hyperparameter tuning, and model evaluation.
- Delivered findings in both written and oral formats to academic examiners, demonstrating the ability to communicate technical work to a specialist audience.

SQL Data Mart | MySQL |

- Designed a 20-table snowflake-schema data mart in MySQL, applying normalisation principles and dimensional modelling to structure data for analytical querying.
- Built and maintained queries, views, and data structures supporting multi-dimensional reporting and aggregation.

Habit-Tracking Application | Python and PostgreSQL |

- Built a PostgreSQL-backed application using Python (object-oriented and functional approaches) to create, update, and query habit and progress data.
- Designed the relational schema and wrote Python logic for data validation, storage, and retrieval.

Power BI Upskilling | Power BI |

- In the process of building multiple Interactive Power BI dashboards.
- This is a skill gap identified in job applications.

EDUCATION

BSc Computer Science | IU International University of Applied Sciences | 2023 to 2026

Overall grade 2.5 on the German scale (1.0 highest). Thesis and colloquium: 1.0. Top module grades: Mathematics I (1.0), Introduction to Programming with Python (1.3), Software Engineering Project (1.3), Software Quality Assurance (1.7), Java and Web Development (1.7). Key modules: algorithms and data structures, OOP in Java and Python, database systems, big data and cloud technologies, computer networks, cybersecurity and cryptography, statistics.

Generative AI Mastermind | Outskill | 2026

Practical application of generative AI tools, prompt engineering, and AI-assisted workflows.

National Senior Certificate | Hoërskool Huguenote | 2022

Mathematics 79%, Computer Applications Technology 84%, Physical Sciences 68%. Top 20 academic leaderboard from 2018 to 2022. Prefect and student council member.

Various Courses | Udemy | 2026

MICROSOFT POWER BI: Power BI Certification in 1 Hour

POWER BI: Microsoft Power BI – Corporate Dashboard Bootcamp

ADDITIONAL

Languages: English (C1 Advanced), Afrikaans (home language), Dutch (B1 Intermediate). Driver's licence held. Travel experience in the Netherlands.